

Experiment # 4

Application of acid base titration

Purpose

1. To understand acid base titration
2. To learn some of the acid base titration applications
3. To determine the concentration (standardize) of an unknown solution of NaOH using the primary standard, potassium hydrogen phthalate.

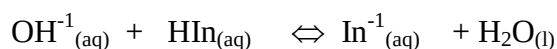
Background

An acid-base **titration** is a procedure used to compare the amount (moles) of acid in one sample with the amount (moles) of base in another. If exactly equivalent amounts of acid or base are used so that neither reactant is present in excess, the solution is said to be at the **equivalence point** 'the point in a titration when a stoichiometric amount of reactant has been added'.

For monoprotic acids and bases (those that react with one H^+ per molecule):

at the equivalence point moles acid = moles base

The indicator is a weak acid (or base) and reacts with some of the titrant to produce the color change. The **endpoint** occurs when all the acid has been neutralized and the solution composition changes suddenly from excess acid to excess base. When the endpoint is reached the addition of titrant should be stopped.



When acid is added to the indicator the **equilibrium shifts** to the left. Because the In^{-1} and Hin have different colors we can see this change. When a base is added to an indicator the hydroxide ions reacts with the molecules of indicator and form indicator ions. The reaction shifts to the right. In this experiment, a phenolphthalein color indicator will be used. Phenolphthalein is colorless in acidic solutions and pink in basic solutions.

Compounds with acidic or basic properties can be determined by titration with either standardized strong acid for basic compounds and strong base for acidic compounds. There are many applications of acid base titration such as calculating the assay of analytical reagent such as strong acids and sodium carbonate in soda ash also we can determined the total acidity of vinegar and total alkalinity of water.

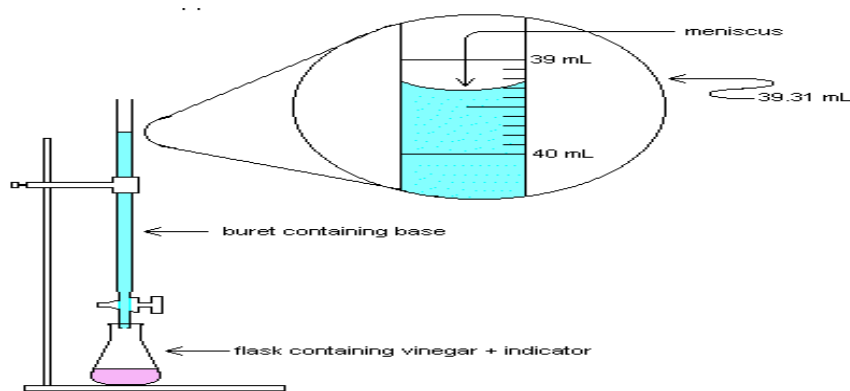
a) Assay of a strong acid:

Strong acid such as nitric, hydrochloric, phosphoric and sulfuric acids are used as raw materials in almost every industry. Assaying their concentrations as wt% or molarity is essential to

determine the amounts required. Usually, certain weight of the acid is diluted to certain volume and a measured portion of the diluted solution is titrated with standardized strong base using a selective indicator.

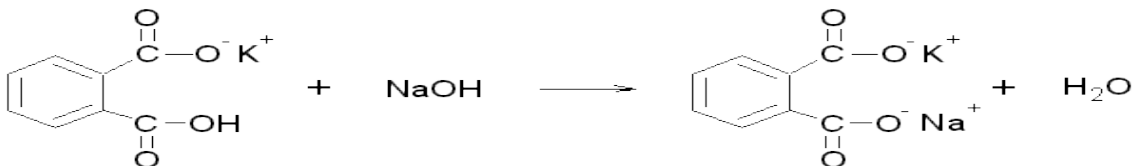
b) Acidity of vinegar:

Vinegar is used in many industrial applications and is usually obtained by fermentation of certain fruits or vegetables. Acidity of vinegar is defined as the total concentration of organic acids in vinegar expressed as percentage of acetic acid. Acidity is an essential parameter in determining the quality of vinegar and has to be assayed before being used. Acidity of vinegar is determined by taking a measured portion, titrating it with strong base (NaOH) using phenolphthalein indicator (ph.ph). The result is expressed as percentage (wt/vol) acetic acid



c) Standardization of NaOH Solution with Potassium Hydrogen Phthalate (KHP)

Sodium hydroxide is hygroscopic and absorbs water from the air when you place it on the balance for weighing. This water will prevent you from being able to find the exact mass of sodium hydroxide. In order to determine the exact concentration of a sodium hydroxide solution you must standardize it by titrating with a solid acid that is not hygroscopic. Potassium hydrogen phthalate, $\text{KHC}_8\text{H}_4\text{O}_4$ (abbreviated KHP), is a non-hygroscopic, crystalline, solid that behaves as a monoprotic acid. It is water soluble and available in high purity. Because of its high purity, you can determine the number of moles of KHP directly from its mass and it is referred to as a primary standard.



You will use this primary standard to **standardize** (determine the exact concentration of a sodium hydroxide solution) by measuring accurately how many milliliters of **titrant** (NaOH solution) are required to exactly neutralize a known amount of acid.

A buret filled with the **titrant** (NaOH solution) is used to measure the volume of NaOH solution added to the known amount of acid in a flask. Phenolphthalein indicator is added to detect the **endpoint**.

Procedure

A) Standardization of NaOH Solution

1. Clean a 250 mL Erlenmeyer flask and then rinse with distilled water.
2. Weigh between 0.3-0.4 grams of potassium hydrogen phthalate (KHP) (204 g/mol) to the nearest 0.0001 g on a piece of weighing paper. Record the exact mass.
3. Transfer the KHP to the 250 mL flask and add about 30 mL of distilled water and swirl to dissolve. The solids must be completely dissolved. Add 2-3 drops of phenolphthalein indicator.
4. Re-fill the burets with the appropriate solution. Record the initial volume readings and titrate the sample to the faint pink end-point. Record the final volume readings.
5. Repeat the procedure for other trials.
6. Calculate the molarity of the NaOH solution for each trial and average these results.

B) Determination of the purity of strong acid

1. Obtain an unknown solid acid and record its number.
2. Weigh between 0.8-1.0 grams of the unknown on the analytical balance.
3. Dissolve the sample in about 50 mL of distilled water and add 2-3 drops of indicator.
4. Titrate the sample with standardized NaOH solution to the first permanent appearance of pink color.
5. Record the volume and repeat the procedure for other trials.
6. Calculate the mass percent of acid for each trial and average the results.

C) Determination of total acidity of vinegar:

1. Ask your instructor to assign to you an unknown acid, write its number
2. Pipet 25.0 mL portion into a 250 mL volumetric flask.
3. Add 2-3 drops of ph.ph. indicator and titrate with standardized NaOH solution to the first permanent appearance of pink color.
4. Record the volume on the data sheet, and calculate the total acidity of vinegar as gram acetic acid per 100 ml of vinegar solution.

Experiment # 4

Application of acid base titration

Report sheet

Name: _____ Number: _____

1. Standardization of a NaOH Solution with Potassium Hydrogen Phthalate (KHP)

	Trial 1	Trial 2	Trial 3
Mass of KHP (g)			
Volume of NaOH used (L)			
Moles of KHP			
Moles of NaOH			
Molarity of NaOH			
Average Molarity of NaOH			

2. Determination of the purity of strong acid

Unknown no.:	Trial 1	Trial 2	Trial 3
Mass of unknown sample (g)			
Molarity of NaOH used			
Volume of NaOH used (L)			
Moles of NaOH			
Moles of unknown acid			
Mass of acid (g)			
Mass percent (%)			
Average Mass percent			

3. Determination of total acidity of vinegar:

Unknown no.:	Trial 1	Trial 2	Trial 3
Volume of unknown vinegar (ml)			
Molarity of NaOH used			
Volume of NaOH used (L)			
Moles of NaOH			
Moles of acetic acid			
Mass of acetic acid (g)			
total acidity of vinegar (g acetic acid/ 100 ml vinegar)			
Average total acidity			

Experiment # 5

Application of acid base titration

Purpose

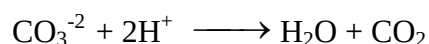
4. To understand acid base titration
5. To learn some of the acid base titration applications

Background

Compounds with acidic or basic properties can be determined by titration with either standardization strong acid for basic compounds and strong base for acidic compounds. There are many applications of acid base titration such as calculating the assay of analytical reagent such as sodium carbonate in soda ash also we can determine the total alkalinity of water.

A) Assay of sodium carbonate in soda ash:

Soda ash is considered one of the major starting materials in inorganic industries, especially glass and ceramics. Soda ash is usually prepared from calcium carbonate and sodium salts. Assay of sodium carbonate content in soda ash is determined by titration of the carbonate content with standardized strong acid using methyl orange indicator with the stoichiometry:



Results are usually reported as % Na_2CO_3 and % Na_2O .

b) Alkalinity of water:

Alkalinity of water is its quantitative capacity to neutralize a strong acid to pH = 4.5 (methyl orange convert from yellow to orange at the end point). Alkalinity is a measure of gross property of water and can be interpreted in terms of specific substance only when the chemical composition is known. Alkalinity is primarily comes from carbonates, bicarbonates and hydroxides of dissolved elements and mainly from bicarbonates for drinking water. Alkalinity is expressed as mg/L CaCO_3 . Alkalinity of water is determined by titrating a measure portion of water with a standardized strong acid using methyl orange indicator assuming the stoichiometry:



Materials

- Standardized hydrochloric acid (HCl) solution
- Soda ash samples
- Tap water samples
- Methyl orange indicator solution
- Phenolphthalein indicator solution

Procedure

Standardization of HCl

- 1- Dissolve 0.20 - 0.25 g sample of previously dried Na_2CO_3 in about 30 ml distilled water.
- 2 - Add 3 drops of methyl orange indicator and titrate the **yellow** solution with standardized HCl until **orange** color is reached.
- 3 - Calculate the molarity of HCl, repeat the procedure and take the average of the results.

Assay of sodium carbonate in soda ash:

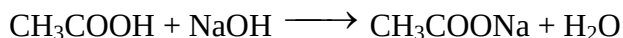
- 1) Ask your instructor to assign you an unknown. Write its number.
- 2) Weigh 1.5-2.0 g of unknown soda ash, dissolve and dilute to 250 mL with distilled water.
- 3) Pipet 25.0 mL from dilute solution into Erlenmeyer flask.
- 4) Add 3 drops of methyl orange indicator and titrate the **yellow** solution with standardized HCl until **orange** color is reached.

Determination of alkalinity of drinking water:

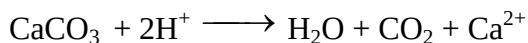
- 1) Pipet 50 mL portion of drinking water into Erlenmeyer flask.
- 2) Add 2-3 drops of methyl orange indicator and titrate to end point. Record volumes

Questions

1. From Part **b**, calculate the alkalinity as mg/L CaCO_3
2. From Part, **a**, calculate the carbonate concentration in the soda ash and express result as % Na_2CO_3 and % Na_2O .
3. A 3.025 g sample of concentrated HNO_3 (63.0 g/mol, 1.12g/mL) were diluted to 250 mL with distilled water. If 25.0 mL aliquot of diluted acid required 17.5 mL of 0.152 M NaOH to reach the ph.ph. End point. What is the wt% of HNO_3 ?
4. A 25.0 mL of grape vinegar was diluted and titrated with 0.2734 M NaOH and required 27.8 mL to reach the ph.ph. End point. What is the acidity of this vinegar % wt/vol acetic acid



5. Calculate the total alkalinity of drinking water if a 50.0 mL aliquot required 20.7 mL of 0.08074 M HCl to reach the methyl orange end point.



6. A 2.00 g sample of juice from a fresh lime was diluted with water and filtered to remove the suspended matter. The clear liquid required 40.39 mL of 0.04022 M NaOH for titration to the ph.ph. End point. Calculate the acidity as % citric acid (194 g/mol)?
7. A 2.5 g sample of soda ash was dissolved and diluted to 250 mL with distilled water. A 25.0 mL aliquot was titrated with 0.150 M HCl and required 12.60 mL for the methyl orange end point. Calculate the % Na_2CO_3 and % Na_2O .
8. What will happen if :
 - a. Soda ash contains some NaHCO_3
 - b. Soda ash contains some NaOH
 - c. Phenolphthalein indicator was used instead of methyl orange

Experiment # 5

Application of acid base titration

Report sheet

Name: _____ Number: _____

1. Standardization of a HCl Solution with Sodium carbonate (Na_2CO_3)

	Trial 1	Trial 2	Trial 3
Mass of Na_2CO_3 (g)			
Moles of Na_2CO_3			
Moles of HCl			
Volume of HCl used (L)			
Molarity of HCl			
Average Molarity of HCl			

2. Determination of the purity of Soda ash

Unknown no.:	Trial 1	Trial 2	Trial 3
Mass of unknown sample (g)			
Molarity of HCl used			
Volume of HCl used (L)			
Moles of HCl			
Moles of Na_2CO_3			
Mass of Na_2CO_3 (g)			
Mass percent (%)			
Average Mass percent			

3. Determination of total alkalinity of water:

	Trial 1	Trial 2	Trial 3
Volume of tap water (ml)			
Molarity of HCl used			
Volume of HCl used (L)			
Moles of HCl			
Moles of CaCO_3			
Mass of CaCO_3 (g)			
total alkalinity of water (ppm CaCO_3)			
Average total acidity			

Experiment 6

Back Titration

Purpose

To determine the concentration of unknown using back titration method

Background

Back titration is a titration done in reverse; instead of titrating the original sample, a known excess of standard reagent is added to the solution, and the excess is titrated. A back titration is useful if the endpoint of the reverse titration is easier to identify than the endpoint of the normal titration.

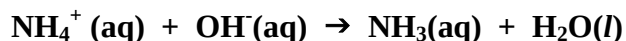
Back titrations are used when:

- a particular reaction is too slow
- an acid or a base is an insoluble salt, for example calcium carbonate
- one of the reactants is volatile, for example ammonia.
- direct titration would involve a weak acid - weak base titration
(the end-point of this type of direct titration is very difficult to observe)

Fertilizer is any organic or inorganic material that is added to soil to supply one or more plant nutrients essential to the growth of plants. Ammonium chloride (NH_4Cl) is an example of inorganic Fertilizer

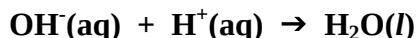
Ammonium ion is acid that is too weak to be titrated directly. However, such a system can be analyzed by a method of "back titration". In this method:

- An excess of standard NaOH is added to react with the ammonium ion.



- This solution is then boiled to evaporate the ammonia.

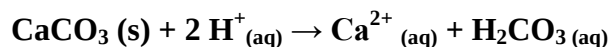
- Finally, the excess hydroxide ion can be titrated with standard HCl, using phenolphthalein as an indicator. This step is called the "back titration."



Antacid are pharmaceutical products that contain bases such as CaCO_3 , MgCO_3 and $\text{Mg}(\text{OH})_2$. Some of them are not very soluble in water and reaction of some of these compounds with HCl is slow. Therefore a back-titration method is used in this experiment. It depends on the addition of

a measured amount of standard HCl to the antacid so that there is a slight excess. After the reaction with antacid is allowed to go to completion, the amount of excess HCl is titrated with standard NaOH using phenolphthalein as indicator.

The active ingredient in antacid is calcium carbonate (CaCO_3). HCl is neutralized by a basic calcium carbonate as:



Procedure

1) Titration of Antacid

1. Accurately weigh antacid tablet (about 0.5 g) and transfer it into 250 mL Erlenmeyer flasks.
2. Add 50 mL of the standardized HCl solution to the sample.
3. Boil each solution gently for 3-5 min and then cool to room temperature (sometimes small amounts of white solid dose not dissolve after heating).
4. Add 3-4 drops of phenolphthalein indicator and titrate each sample with NaOH to obtain permanent pink. Record the volume of the base used.
5. Use the number of moles of HCl neutralized by the tablet to determine the mass of the calcium carbonate in the tablet. The molar mass of CaCO_3 is 100 g/mol.

2) Titration of ammonium chloride

1. On the analytical balance weigh about 0.3 g of the unknown salt into a 250 mL Erlenmeyer flask.
2. Add about 50 mL of the standard NaOH to the sample.
3. Dissolve the salt and boil the solution gently on the hot plate. Check for the presence of NH_3 in the steam coming from the mouth of the flask by holding a piece of moist red litmus paper over the stream of vapor leaving the flask. A blue color indicated that NH_3 is still present in the solution. Continue boiling until a negative test result is obtained (red litmus should remain red).
4. When no more ammonia is present, remove the flask from the hot plate and cool the solution by adding 25 ml of distilled water back into the flask.
6. Add about 5 drops of methyl orange indicator and titrate the excess NaOH with the standardized HCl solution until the color changes from yellow to orange.
7. Calculate the mass percent of the ammonium chloride (53.5 g/ mol) in your unknown.

**Experiment 6 (Back Titration):
Report sheet**

Name: _____ **Number:** _____

Determination of % NH_4Cl in fertilizer

Unknown no.....	Trial 1	Trial 2
Mass of unknown (g)		
Volume of OH^- added (L) Molarity of OH^- Moles of OH^- added		
Volume of HCl back titrated (L) Molarity of HCl Moles of HCl back titrated Moles of NaOH unreacted		
Moles of NaOH reacted Moles of NH_4Cl Mass of NH_4Cl % NH_4Cl		
Average % NH_4Cl		

Determination of % CaCO_3 in Antacid

Mass of Antacid tablet	Trial 1	Trial 2
Volume of HCl added (L) Molarity of HCl Moles of HCl		
Volume of NaOH back titrated Molarity of NaOH Moles of NaOH back titrated		
Mol CaCO_3 Mass of CaCO_3 (g) % CaCO_3		
Average % CaCO_3		

